3 A fixed mass of gas undergoes a cycle of changes of pressure, volume and temperature as shown in Fig. 3.1. The gas is assumed to be ideal.

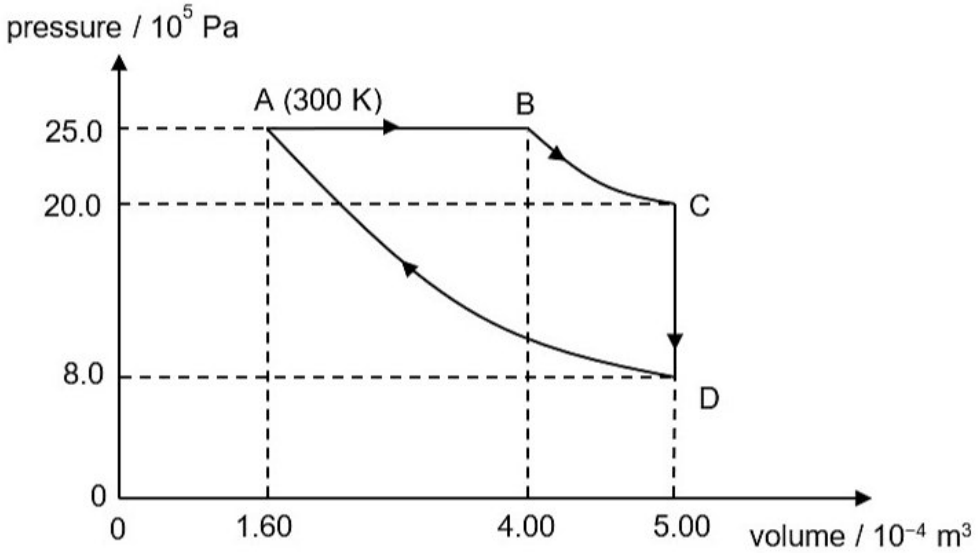


Fig. 3.1 (Not to scale)

The temperature of the gas at A is 300 K.

(a) Show that the amount of gas is 0.160 mol.

[1]

(b) Calculate the temperature at B.

temperature = ... K [1]

Fig. 3.2 indicates the type of the various processes in the cycle:

Process	Type of process
A to B	isobaric (constant pressure)
B to C	isothermal (constant temperature)
C to D	isochoric (constant volume)

Fig. 3.2

(c) Explain why process D to A is likely an isothermal process.

.....

.....

.....

.....

..... [2]

Fig. 3.3 lists some data about the gas:

Quantity	Value
molar mass	0.0200 kg mol^{-1}
energy transferred to the gas from A to B by heating	1500 J
specific heat capacity at constant volume	625 J kg^{-1} K^{-1}

Fig. 3.3

(d) (i) Using the data from Fig. 3.3, show that the specific heat capacity of the gas from A to B at constant pressure is 1040 J kg^{-1} K^{-1}.

[1]

(ii) Use the first law of thermodynamics to determine the increase in internal energy ΔU of the gas from A to B.

ΔU = ... J [2]

(iii) Using the data from Fig. 3.3, calculate the heat flow during the process C to D.

heat flow during the process C to D = ... J [1]

(e) Fig. 3.4 shows the increase in internal energy ΔU of the gas, the heat transferred q to the gas by heating, and the work done on the gas w during the various processes of the cycle. During B to C, 220 J of energy is transferred to the gas by heating and during D to A, work done on the gas is 320 J.

Process	ΔU / J	q / J	w / J
A to B			
B to C		**220**	
C to D			
D to A			**320**

Fig. 3.4

Complete the table in Fig. 3.4. [4]

[Total: 12]